Assignment operation – Important topic in JavaScript, Java, C# …

Applies to all main programming languages, not only JavaScript.

JavaScript examples with beige or tan colored boxes. Other language examples with blue or other colors

```
let a;   // Variable ‘a’ is defined/created. No value yet => value is still ”undefined”
a = 3;   // Left side tells where to save. Right side tells which value to save there.
```

Write down and memorize these simple rules:

- If you see a **single equals mark alone** between two items = it means the assignment operation. Look above.
- **Left side** ALWAYS tells the **location where to save**, and only that.
- **Right side** ALWAYS tells the value which will be saved to the location on left. Right side might have more complicated evaluations, but finally it must produce **the value to be saved.**

```
let name = document.getElementById(”txtName”).value;
// Where to save? To the variable called ‘name’
// What to save? The value from the txtName input field of the HTML document=web page=DOM.
```

In some languages they use different symbol for assignment

JavaScript, Java, C, Python and many other programming languages you use the equal sign, = , for assignment. But some other languages use different symbols that may be clearer, more understandable and logical, for example:

```
// These are examples of assignment in few other languages. They are NOT JavaScript.

a := 3;   // This is how you assign value 3 to variable ‘a’ e.g. in ALGOL and Pascal.

a << 3;   // This is how you assign value 3 to variable ‘a’ e.g. in Magik language.

a <- 3;   // This is how you assign value 3 to variable ‘a’ e.g. in F# and R languages.
```

Would it be easier for you to learn what **a = b** means, if it was written **a <- b**, for example?

```
let a = 3;  // Variable ‘a’ defined (created) and value 3 assigned to it.
let b = 7;  // Variable ‘b’ defined (created) and value 7 assigned to it.
b = a;      // What’s the value of ‘a’ now?
a = 22;     // What’s the value of ‘a’ now? What’s the value of ‘b’ now? (Answers: 22 and 3)
```

If you thought ‘b’ would have the value 22 you forgot the rules from the start. Recap the rules in any case.

Even if the statements before **and/or** after the assignment operator ‘=’ would look complicated, the same simple rules do still always rule: **left side tells where to save** and **right side tells what/which value to save**.

```
document.getElementById(”txtWinner”).value = document.getElementById(”txtName1”).value;
```

Here the value from element with id ‘txtName1’ is saved to the ‘value’ property of the element with the id ‘txtWinner’.